

JEE-Main-24-01-2026 (Memory Based)

[MORNING SHIFT]

Physics

Question: A spring of stiffness $k = 15 \text{ N/m}$ is cut into a ratio of $3 : 1$. Find the spring constant of smaller length spring thus formed.

Options:

- (a) 15 N/m
- (b) 30 N/m
- (c) 45 N/m
- (d) 60 N/m

Answer: (d)

Solution:

Soln. We know $k \propto \frac{1}{l}$
 $\frac{k_1}{k_2} = \frac{l_2}{l_1}$
 $\therefore k_1 : k_2 = 1 : 3$
 $k_1 = 15 \text{ N/m}$ $k_2 = ?$
 $\frac{15}{k_2} = \frac{1}{3}$
 $k_2 = 15 \times 3 = 45 \text{ N/m}$
 Equivalent $k = \frac{k_1 k_2}{k_1 + k_2}$
 $15 = \frac{x \times 3x}{4x}$
 $15 = \frac{3x}{4}$
 $15 \times 4 = 3x$
 $60 = 3x$
 $x = 20$
 Hence k_2 is 60 N/m

Question: EM waves and their source are given

	Column-I		Column-II
a)	X-rays	p)	Hot bodies and molecules
b)	Infrared Rays	q)	Oscillatory current in Antennas
c)	Microwaves	r)	Magnetron
d)	Radio waves	s)	Fast moving electrons striking a metal plate

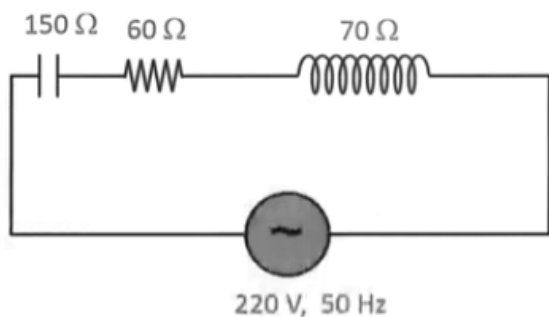
Options:

- (a) a-p, b-s, c-r, d-q
- (b) a-s, b-p, c-r, d-q
- (c) a-s, b-p, c-s, d-q
- (d) a-s, b-r, c-p, d-q

Answer: (b)

Solution:

Question: For the given circuit, find the power factor.



Options:

- (a) 4/5
- (b) 3/5
- (c) 3/4
- (d) 4/3

Answer: (b)

Solution:

Solution:- Power factor, $\cos \phi = \frac{R}{Z} = \frac{360}{100} = \frac{3}{5}$

$$Z = \sqrt{60^2 + (150 - 70)^2} = 100$$

$\cos \phi = \frac{3}{5} \quad (b)$

Question: In H-like atom ratio of speed in two orbits is 3 : 2, then ratio of energy is

Options:

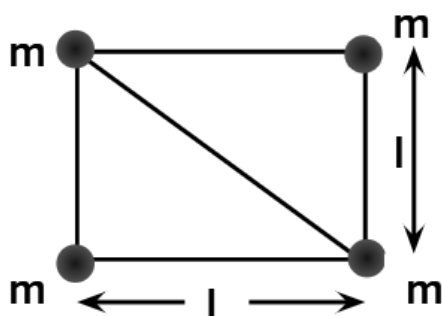
- (a) 2 : 3
- (b) 9 : 4
- (c) 2 : 1
- (d) 5 : 3

Answer: (b)

Solution:

$$\frac{v_1}{v_2} \propto \frac{Z_1/n_1}{Z_2/n_2} \quad \frac{E_1}{E_2} = \frac{Z_1^2/n_1^2}{Z_2^2/n_2^2}$$

Question: 4 particles each of mass m are present on the 4 radius of a sequence of length 'l'. Find the moment of inertia of the system about an axis passing through one of the vertices & perpendicular to the plane of the square.

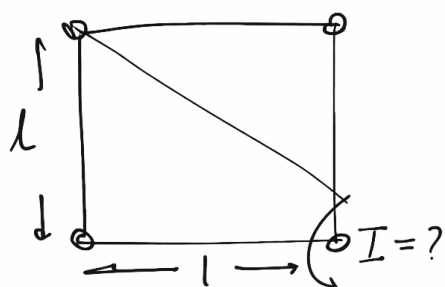


Options:

- (a) ml^2
- (b) $3ml^2$
- (c) $4ml^2$
- (d) $6ml^2$

Answer: (c)

Solution:



$$I = 0 + ml^2 + m(\sqrt{5}l)^2 + mn l^2$$

$$= 4ml^2$$

Question: Find the de broglie wavelength for O_2 molecules. Who are maintained at a temperature of $27^\circ C$.

(Mass of $O_2 = 5.3 \times 10^{-26}$ kg)

Options:

- (a) $2.3 \text{ \AA}$
- (b) $6.1 \text{ \AA}$
- (c) $5.2 \text{ \AA}$
- (d) $8 \text{ \AA}$

Answer: (a)

Solution:

$$K.E = \frac{3}{2} kT$$

$$\frac{p^2}{2m} = \frac{3}{2} kT$$

$$p = \sqrt{3mkT}$$

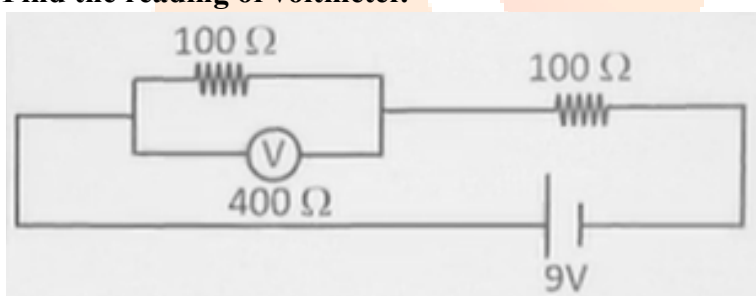
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3mkT}}$$

$$= \frac{6.6 \times 10^{-34}}{\sqrt{3 \times 5.31 \times 10^{-26} \times 1.4 \times 10^{-23} \times (27+273)}}$$

$$= \frac{6.6 \times 10^{-34}}{9 \times 10^{-24} \sqrt{151}}$$

$$= \frac{6.6 \times 10^{-34}}{9 \times 0.32}$$

Question: A voltmeter of 400Ω resistance is in parallel with 100Ω resistor. And the combination is connected with 100Ω resistor and a battery of 9 volt in series as shown. Find the reading of voltmeter.



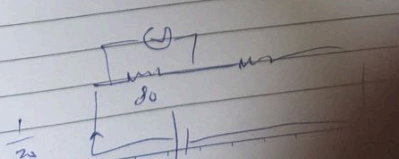
Options:

- (a) 5 volts
- (b) 3 volts
- (c) 4 volts
- (d) 6 volts

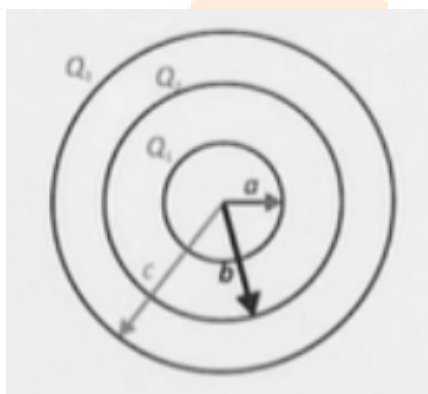
Answer: (c)

Solution:

Solⁿ Total Resistance = $\frac{100 \times 20}{50} + 100$
 $= 80 + 100$
 $= 180$
 $i = \frac{V}{R} = \frac{9}{180} = \frac{1}{20}$
 $V = R \times i = 80 \times \frac{1}{20} = 4V$



Question: Three uniformly concentric charged shells are kept as shown. Find potential of each shell.



Options:

- (a) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$, $V_B = \frac{k(Q_1 + Q_2 + Q_3)}{b}$, $V_C = \frac{k(Q_1 + Q_2 + Q_3)}{c}$
- (b) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$, $V_B = \frac{k(Q_1 + Q_2)}{b} + \frac{kQ_3}{c}$, $V_C = \frac{k(Q_1 + Q_2 + Q_3)}{c}$
- (c) $V_A = \frac{kQ_1}{a} + \frac{k(Q_2 + Q_3)}{c}$, $V_B = \frac{k(Q_1 + Q_2)}{b} + \frac{kQ_3}{c}$, $V_C = \frac{k(Q_1 + Q_2 + Q_3)}{c}$
- (d) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$, $V_B = \frac{k(Q_1 + Q_2)}{a} + \frac{kQ_3}{b}$, $V_C = \frac{k(Q_1 + Q_2 + Q_3)}{c}$

Answer: (b)

Solution:

$$V_a = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$$

$$V_b = \frac{kQ_1}{b} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$$

$$V_c = \frac{kQ_3}{c} + \frac{kQ_1}{c} + \frac{kQ_2}{c}$$

Question: Find the ratio of maximum wavelength in Lyman series to that of maximum wavelength in Balmer series in Hydrogen Spectrum?

Options:

- (a) 27/5
- (b) 5/27
- (c) 3/4
- (d) 9/5

Answer: (b)

Solution:

Handwritten solution showing the calculation of the ratio of maximum wavelengths in the Lyman and Balmer series.

For Lyman:

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$\frac{1}{\lambda_1} = R(1)^2 \left[1 - \frac{1}{(\infty)^2} \right] = \frac{3R}{4}$$

$$\lambda_1 = \frac{4}{3R}$$

For Balmer:

$$\frac{1}{\lambda_2} = R(1)^2 \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{5R}{36}$$

$$\lambda_2 = \frac{36}{5R}$$

$\therefore \frac{(\lambda_{\max})_{\text{Lyman}}}{(\lambda_{\max})_{\text{Balmer}}} = \frac{\lambda_1}{\lambda_2} = \frac{4}{3} \times \frac{5}{36} = \frac{5}{27}$

Question: A conducting rod of length 10 cm is dropped vertically from rest through a height 10 m in a region of uniform inward magnetic field of magnitude 0.5T. The rod is always remains Perpendicular to magnetic field. Neglect air resistance and electrical resistance. The magnitude of induced emf in the rod just before it reaches ground is

Options:

- (a) $0.25\sqrt{2}$ V
- (b) 0.5 V
- (c) $0.5\sqrt{2}$ V
- (d) $\sqrt{2}$ V

Answer: (c)

Solution:

$$v = \sqrt{2gh}$$

$$= \sqrt{2 \times 10 \times 10}$$

$$v = 10\sqrt{2} \text{ m/s}$$

$$e = Blv$$

$$= 0.5 \times 10 \times 10^{-2} \times 10\sqrt{2}$$

$$e = 0.5\sqrt{2} \text{ V}$$

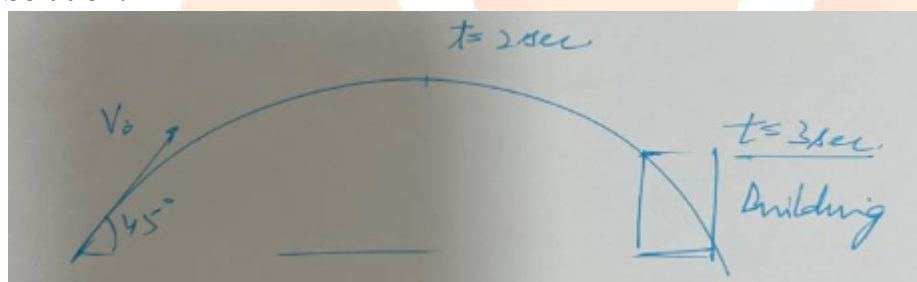
Question: A particle thrown at an angle of 45° . It reaches its highest point of trajectory after $t = 2$ sec. It hits a building located on the ground at some distance from its point of projection after 3 sec. Find the height of the building.

Options:

- (a) 30 m
- (b) 15 m
- (c) 45 m
- (d) None of these

Answer: (b)

Solution:



$$T = \frac{2u \sin \theta}{g} \quad \therefore 4 = \frac{2u \times \frac{1}{\sqrt{2}}}{g}$$

$$\therefore u_0 = 20\sqrt{2} \text{ m/s}$$

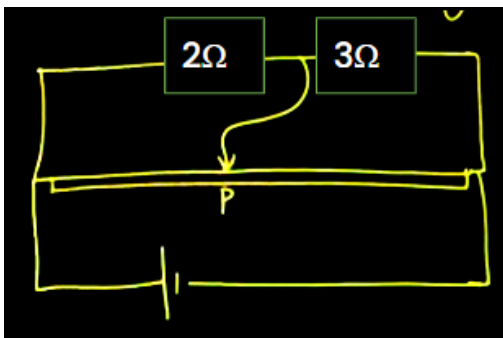
The height of the particle after $t = 3 \text{ sec.}$

$$s = ut + \frac{1}{2}at^2$$

$$h = 20 \times 3 + \frac{1}{2} \times (-10) \times (3)^2$$

$$= 60 - 45 = 15 \text{ m.}$$

Question: Figure shows a meter bridge. Initially null points was achieved at P. When an unknown resistance R is connected in parallel with 3Ω then the point is shifted by 22.5 cm. Find the unknown R.

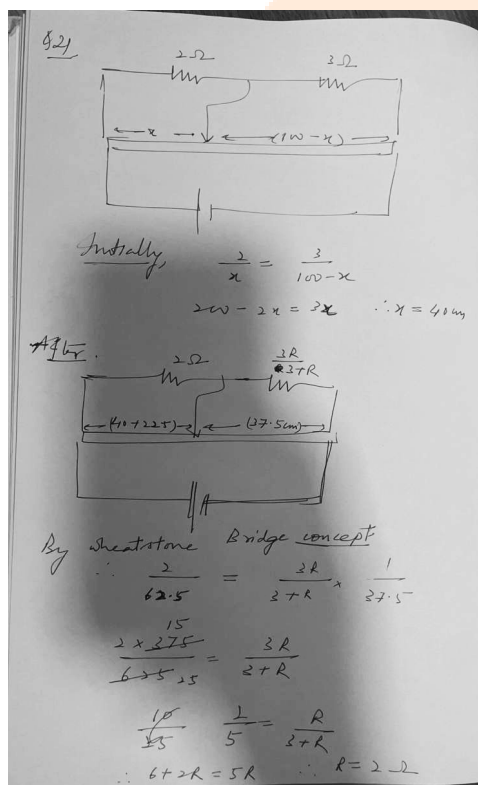


Options:

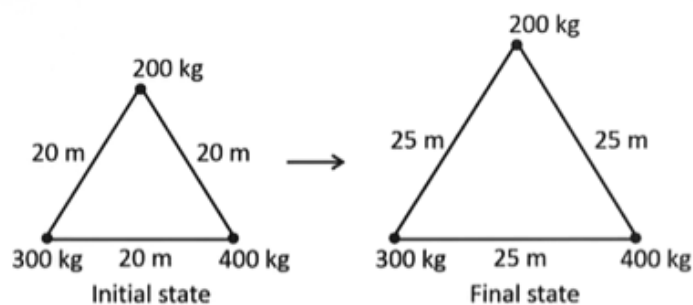
- (a) 2Ω
- (b) 3Ω
- (c) 2.5Ω
- (d) 5Ω

Answer: (a)

Solution:



Question: There are 3 particles present at the vertex of an equilateral triangle of length 20 cm. Find the work done to increase the distance between them from 20 cm to 25 cm.



Options:

- (a) $1.7342 \times 10^{-7} \text{ J}$
- (b) $1.6253 \times 10^{-7} \text{ J}$
- (c) $2.5232 \times 10^{-7} \text{ J}$
- (d) $6.6325 \times 10^{-7} \text{ J}$

Answer: (a)

Solution:

$$U_i = -\frac{G}{20} \left[\frac{200 \times 300}{1} + \frac{300 \times 400}{1} + \frac{200 \times 400}{1} \right]$$

$$U_f = -\frac{G}{25} \left[\frac{200 \times 300}{1} + \frac{300 \times 400}{1} + \frac{200 \times 400}{1} \right]$$

$$= \left(\frac{4}{5} - 1 \right) U_i$$

$$\Delta W = 1.7342 \times 10^{-7} \text{ J}$$

Question: Electric potential at a point is given as $E = Ar^3 + B$. Find charge in sphere of radius 1 m.

Options:

- (a) $-4\epsilon_0 A\pi$
- (b) $-8\epsilon_0 A\pi$
- (c) $-12\epsilon_0 A\pi$
- (d) $-16\epsilon_0 A\pi$

Answer: (c)

Solution:

Solution:

$$E = -\frac{dv}{dr} = -\frac{d}{dr}(Ar^3 + B)$$

$$= -3Ar^2 \quad [r=1]$$

$$\boxed{E = -3A}$$

By Gauss law, $E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$

$$-3A \cdot 4\pi (1)^2 = \frac{Q}{\epsilon_0}$$

$$\boxed{-12\pi A \epsilon_0 = Q} \text{ Ans}$$

Question: A light wave is incident from medium 1 to medium 2 speed of light in medium 1 is $V_1 = 2.4 \times 10^8$ m/s, & speed in medium 2 is $v_2 = 2.7 \times 10^8$ m/s. Find the angle.

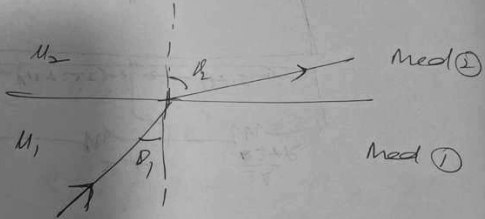
Options:

- (a) $\sin^{-1}(0.88)$
- (b) $\tan^{-1}(0.88)$
- (c) $\cos^{-1}(0.8)$
- (d) $\sec^{-1}(0.8)$

Answer: (a)

Solution:

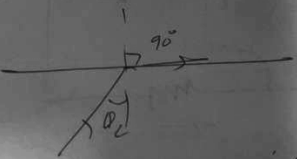
Ans.



$$\mu_1 = \frac{c}{v_1} = \frac{3 \times 10^8}{2.4 \times 10^8} = 1.25$$

$$\mu_2 = \frac{c}{v_2} = \frac{3 \times 10^8}{2.7 \times 10^8} = \frac{10}{9} = 1.11$$

For critical angle.



$$\mu_1 \sin \theta_c = \mu_2$$

$$\sin \theta_c = \frac{1.11}{1.25} = 0.88$$

$$\therefore \theta_c = \sin^{-1}(0.88)$$

Question: Two thin prisms are combined in such a way that there is deviation without dispersion. If $H_v = 1.523$ & $H_r = 1.515$ are refractive index for P_1 and $H_v = 1.666$ & $H_r = 1.650$ for P_2 . Then find the angle of prism for P_2 . (Given $A_1 = 10^\circ$)

Options:

- (a) 5°
- (b) 7.8°
- (c) 10.5°
- (d) 20°

Answer: (c)

Solution:

Handwritten solution showing the calculation for the angle of prism A_2 for prism P_2 . The formula used is $A_2 = \frac{(H_v - H_r) A_1}{H_v - H_r}$. Substituting the values, $A_2 = \frac{(1.523 - 1.515) 10^\circ}{1.666 - 1.650}$, which results in $A_2 = 10.5^\circ$.

Question: Match the two Lists given below

	List-I		List-II
a)	Magnetic flux	1)	$M^1 L^2 T^{-2} A^{-2}$
b)	Magnetic permeability	2)	$M^1 L^2 T^{-2} A^{-1}$
c)	Magnetic induction	3)	$M^1 L^1 T^{-2} A^{-2}$
d)	Self induction	4)	$M^1 L^0 T^{-2} A^{-1}$

Options:

- (a) a-2, b-3, c-4, d-1
- (b) a-3, b-2, c-1, d-4
- (c) a-4, b-3, c-1, d-2
- (d) a-1, b-2, c-3, d-4

Answer: (a)

Solution:

$$B = \frac{F}{IL}$$

$$[B] = \frac{M T^{-2} A^{-1}}{L}$$

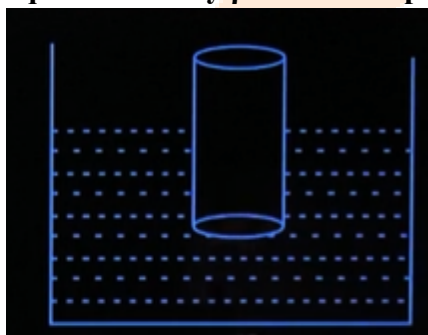
$$[BA] = [\phi] =$$

$$= M L^2 T^{-2} A^{-1}$$

$$\mu_0 = \frac{BL}{I}$$

$$M L T^{-2} A^{-2}$$

Question: A cylinder of mass m , length l and area of cross section A is in equilibrium in liquid of density ρ . Find time period of small vertical oscillations.



Options:

(a) $2\pi \sqrt{\frac{mA}{\rho g}}$

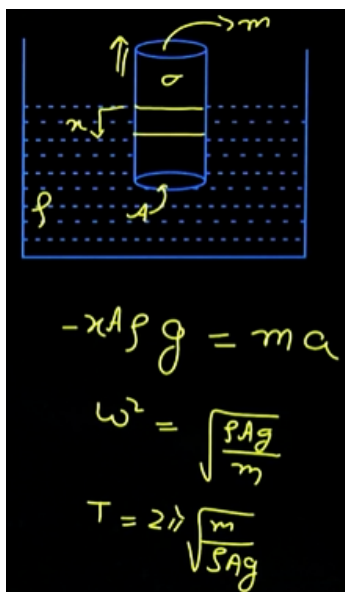
(b) $2\pi \sqrt{\frac{mg}{\rho A}}$

(c) $2\pi \sqrt{\frac{m}{\rho A^2 g}}$

(d) $2\pi \sqrt{\frac{m}{\rho A g}}$

Answer: (d)

Solution:



Question: Match the two Lists given below

	Column-I		Column-II
1)	Faraday's law of EMI	A)	$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$
2)	Ampere's circuital law	B)	$\Phi = \frac{q_{in}}{\epsilon_0}$
3)	Gauss law	C)	$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$
4)	Maxwell's equations (Gauss's law for magnetism)	D)	$\Delta \vec{B} = 0$

Options:

- (a) 1-C, 2-A, 3-B, 4-D
- (b) 1-A, 2-C, 3-D, 4-B
- (c) 1-C, 2-B, 3-A, 4-D
- (d) 1-D, 2-A, 3-B, 4-C

Answer: (a)

Question: A dipole is placed in uniform magnetic field $B = 800$ gauss at an angle 30° then it experiences the torque of 16×10^{-3} N-m. Find the work done in slowly moving the dipole from stable equilibrium to unstable equilibrium.

Options:

- (a) 64×10^{-3} J
- (b) 5×10^{-3} J
- (c) 24.5×10^{-3} J
- (d) 7.6×10^{-3} J

Answer: (a)

Solution:

$$\begin{aligned}
 & \xrightarrow{M} B \quad U_i = -MB \quad \tau = MB \sin 30^\circ \\
 & 16 \text{ mN} = \frac{MB}{2} \quad \text{--- (1)} \\
 & \xleftarrow{M} B \\
 & U_f = +MB \\
 & \Delta U = W_{\text{ext}} = 2MB = 64 \text{ mJ}
 \end{aligned}$$

Question: A brass rod is rigidly fixed and maintained at 27°C . If it cools down to -43°C then the tension in the rod is T_0 . Find at what temperature tension will be $1.4 T_0$.

Options:

- (a) -71°C
- (b) 108°C
- (c) 71°C
- (d) -108°C

Answer: (a)

Solution:

$$\begin{aligned}
 & \frac{1}{1.4} = \frac{-70}{21-27} \\
 & 21-27 = -70 \times 1.4 \\
 & \quad = -70 \times 1.4 + 27 \\
 & 21 = -71^\circ\text{C}
 \end{aligned}$$

Question: In an screw gauge if 7th division is coinciding with the horizontal main state when nothing is there than find the actual diameter of pen when main scale reading is 7 and circular division is 65th.

Options:

- (a) 7.58 mm
- (b) 7.72 mm
- (c) 7.65 mm
- (d) None of these

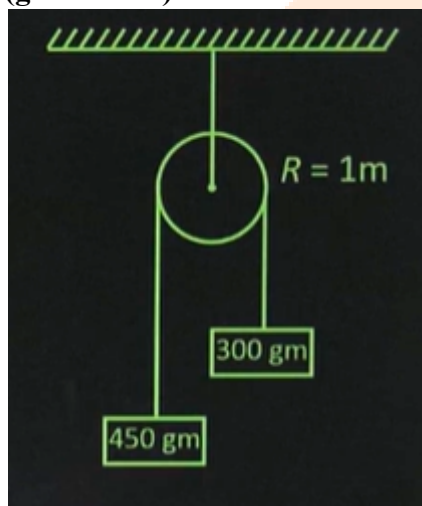
Answer: (a)

Solution:

$$\begin{aligned} \text{Zero Error} &: 7 \times 0.01 \text{ mm} \\ \text{Actual Reading} &: 7 + 65 \times 0.01 - \text{Zero error} \\ &: 7 + 0.65 - 0.07 \\ &: 7.58 \text{ mm} \end{aligned}$$

Question: When system is released from rest the heavier mass goes 81 cm in 9 sec, find rotational inertia.

($g = 10 \text{ m/s}^2$)



Options:

- (a) 74.25 kg-m^2
- (b) 100.25 kg-m^2
- (c) 50.25 kg-m^2
- (d) 25.25 kg-m^2

Answer: (a)

Solution:

$$\frac{1}{2} a t^2 = 0.81$$

$$a = 0.02 \text{ m/s}^2$$

$$a = \frac{(m_2 - m_1) g}{m_1 + m_2 + \frac{I}{R^2}}$$

$$0.02 = \frac{0.15 \times 10}{0.75 + \frac{I}{10^2}}$$

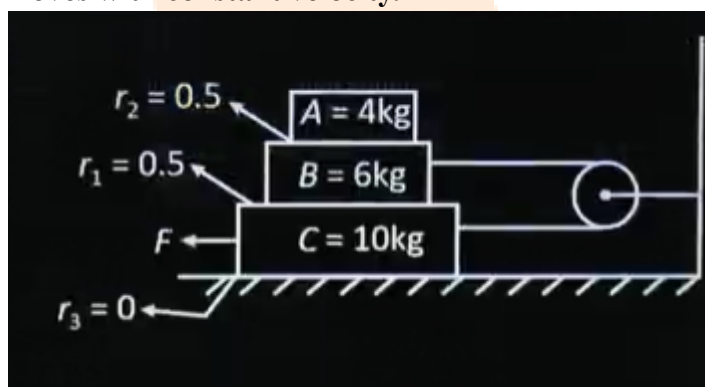
Question: Statement-1: Binding energy per nucleon always increase with mass number.
Statement-2: Binding energy per nucleon for smaller mass number always performs nucleon fusion.

Options:

- (a) Statement-1, true Statement-2, false
- (b) Statement-1, true Statement-2, true
- (c) Statement-1, false Statement-2, true
- (d) Statement-1, false Statement-2, false

Answer: (c)

Question: For the given arrangement find the value of F(in Newton) so that body c moves with constant velocity.

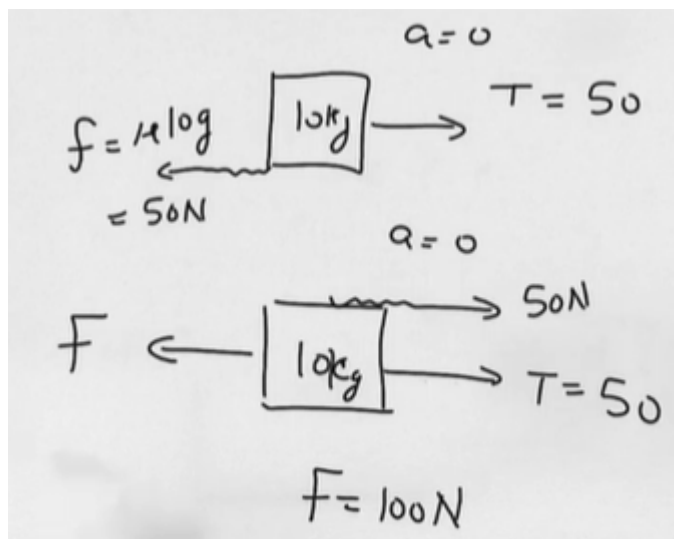


Options:

- (a) 100 N
- (b) 200 N
- (c) 300 N
- (d) 400 N

Answer: (100 N)

Solution:



Question: In YDSE the wavelengths of lights used are λ_1 and λ_2 . The separation between slits are d_1 and d_2 . Distance between slit and screen are D_1 and D_2 Respectively. If fringe

width obtained in both experiments are same then ratio of $\frac{D_1}{D_2}$ is

Options:

- (a) $\frac{\lambda_1 d_1}{\lambda_2 d_2}$
- (b) $\frac{\lambda_1 d_2}{\lambda_1 d_1}$
- (c) $\frac{\lambda_2 d_1}{\lambda_2 d_2}$
- (d) $\frac{\lambda_1 d_1}{\lambda_1 d_2}$

Answer: (b)

Solution:

Ans: $P_1 = P_2$

$$\frac{\lambda_1 D_1}{d_1} = \frac{\lambda_2 D_2}{d_2}$$

$$\boxed{\frac{D_1}{D_2} = \frac{\lambda_2 d_1}{\lambda_1 d_2}}$$

Question: In a cylindrical flux two lens of focal length 25 cm & 5 cm are arranged in such a way that the can object a very for placed object with a magnification of 5^k . Find value of k.

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) -4

Answer: (a)

Solution:

It's a telescope cause we are seeing far placed object

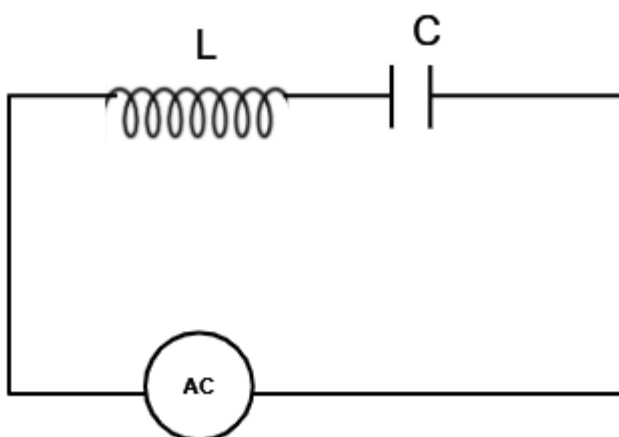
$$m = \frac{f_o}{f_e}$$

$$5^k = \frac{25}{5}$$

$$5^k = 5^1$$

$$k = 1$$

Question: In the given LC circuit, EMF of the circuit is given by $E_0 \sin \omega t$. If capacitance of capacitor is C and inductance is L then find the value of max current.



Options:

- (a) $V_0 \sqrt{\frac{C}{L}}$
- (b) $\frac{V_0}{\sqrt{LC}}$
- (c) $\sqrt{C}$
- (d) None of these

Answer: (a)

Solution:

Handwritten solution on lined paper:

Max Charge on Capacitor $Q_0 = C \times V_0$

Max Current: $i_0 = \frac{Q_0}{\sqrt{LC}}$

$= \frac{C V_0}{\sqrt{LC}}$

$= \sqrt{\frac{C}{L}} V_0$

Question: Velocity of electron in n^{th} shell of a hydrogen like atom is $3 \times 10^5 \text{ m/s}$ and velocity of electron in m^{th} shell of that atom is $2.5 \times 10^5 \text{ m/s}$. Find ratio of radius of m^{th} shell to n^{th} shell.

Options:

- (a) 25/40
- (b) 25/36
- (c) 36/25
- (d) 36/35

Answer: (c)

Solution:

$$v_n = \frac{v_0}{n}$$

$$r_n = r_0 \frac{n^2}{Z}$$

$$\frac{v_n}{v_m} = \frac{m}{n} = \frac{6}{5}$$

$$\frac{r_m}{r_n} = \frac{m^2}{n^2} = \frac{36}{25}$$

Question: Four lengths l_1, l_2, l_3, l_4 are measured independently. These lengths are added to obtain total length, $L = l_1 + l_2 + l_3 + l_4$. Each length is measured with same absolute error Δl . Find $\Delta L/L$

Options:

- (a) $\frac{\Delta l}{l_1 + l_2 + l_3 + l_4}$
- (b) $\frac{4\Delta l}{l_1 + l_2 + l_3 + l_4}$
- (c) $\frac{\Delta l}{l_1 l_2 l_3 l_4}$
- (d) $\frac{4(l_1 + l_2 + l_3 + l_4)}{\Delta l}$

Answer: (b)

Solution:

Ans $\Delta L = \Delta l_1 + \Delta l_2 + \Delta l_3 + \Delta l_4$

$$\Delta L = 4\Delta l$$

$$\therefore \frac{\Delta L}{L} = \frac{4\Delta l}{l_1 + l_2 + l_3 + l_4}$$

Question: The time period and length of the Pendulum is given as $T_1 = 25$, $T_2 = 2T_1$, $l_1 = 50$ cm. Find l_2 .

Options:

- (a) 100 cm

- (b) 150 cm
- (c) 200 cm
- (d) 50 cm

Answer: (c)

Solution:

$$\text{Ans: } T = 2\pi \sqrt{\frac{l}{g}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}}$$

$$\frac{1}{2} = \sqrt{\frac{50}{l_2}}$$

$$\frac{1}{4} = \frac{50}{l_2} \Rightarrow \boxed{l_2 = 200 \text{ cm}}$$

Question: For a microscope focal length of objective is 2 cm and focal length of eyepiece is 4 cm. Tube length is $L = 10$ cm. Magnification for normal adjustment is $5x$. Find the value of x .

Options:

- (a) 2
- (b) 3
- (c) 4
- (d) 6

Answer: (a)

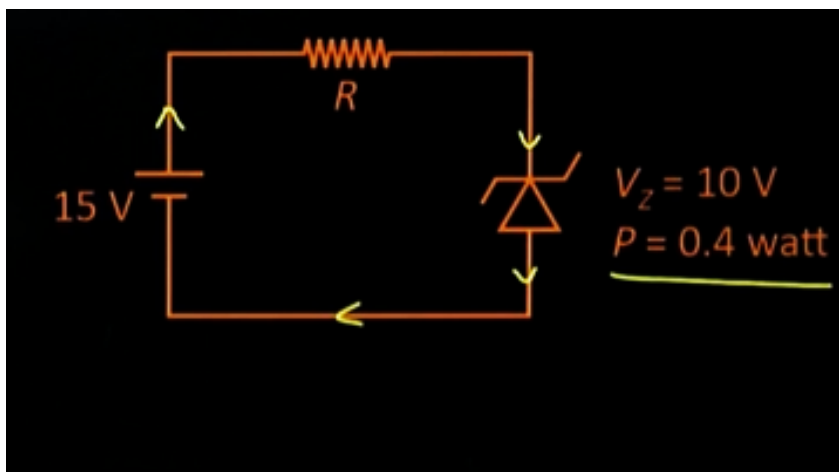
Solution:

$$M = \frac{L D}{f_o f_e} = 5^x$$

$$= \frac{10 \times 25}{2 \times 5} = 5^2$$

$$x = 2$$

Question: For the given the breakdown voltage of Zener diode is $V_z = 10$ volts and it can with-stand the power dissipation of 0.4 watt. Find the value of resistance R (in Ω)



Options:

- (a) 120Ω
- (b) 125Ω
- (c) 135Ω
- (d) 110Ω

Answer: (b)

Solution:

$$V = i_{\max} R_{\min} \quad V i = P \Rightarrow i = \frac{4}{100}$$

$$\Rightarrow \frac{5}{4} \times 100 = 125 \Omega = R_{\min}$$

Question: If potential varies as distance r as $v(r) = ar^3 + b$. Total magnitude of charge Q inclosed within a sphere of unit radius is $Q = a(\pi a \epsilon_0)$. Find the value of a .

Options:

- (a) 12
- (b) 16
- (c) 14
- (d) 13

Answer: (a)

Solution:

$$V = ar^3 + b$$

$$E = -\frac{dV}{dr} = -3ar^2$$

Gauss law

$$E(r) 4\pi r^2 = \frac{Q_{in}}{\epsilon_0}$$

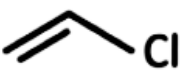
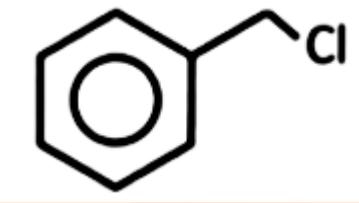
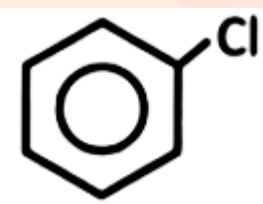
$$12\pi a = \frac{Q_{in}}{\epsilon_0}$$

$$\alpha = 12$$



JEE-Main-24-01-2026 (Memory Based)
[MORNING SHIFT]
Chemistry

Question: Match List-I with List-II

List-I	List-II
A) Vinyl halide	i) 
B) Allyl halide	ii) 
C) Benzyl halide	iii) 
D) Aryl halide	iv) 

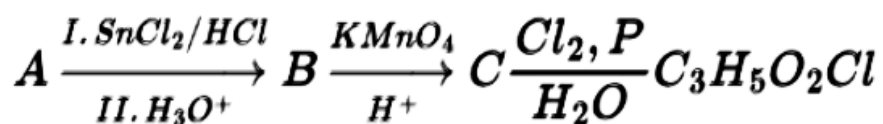
Select the correct option.

Options:

- (a) A(ii), B(i), C(iii), D(iv)
- (b) A(i), B(ii), C(iii), D(iv)
- (c) A(i), B(ii), C(iv), D(iii)
- (d) A(ii), B(i), C(v), D(iii)

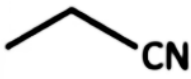
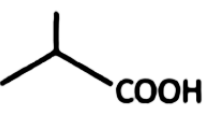
Answer: (b)

Question:



Final product has one chiral centre. Structure of A is

Options:

- (a) 
- (b) 
- (c) 
- (d) 

Answer: (a)

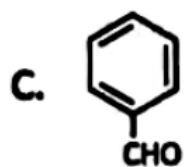
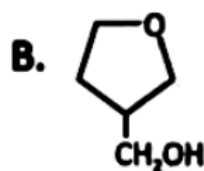
Question: Which of following compound contains 3 unpaired electrons?

Options:

- (a) V_2O_5
 (b) $[TiF_6]^{3-}$
 (c) $[CoF_6]^{4-}$
 (d) $[Fe(CN)_6]^{3-}$

Answer: (c)

Question: Which of the following compounds with give positive Tollen's reagent test?

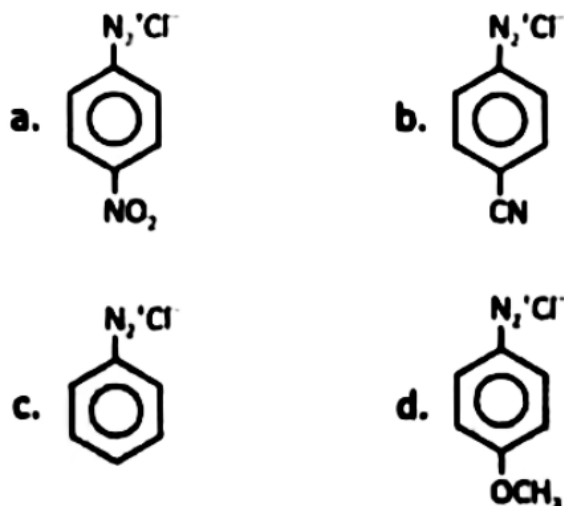


Options:

- (a) A, B and C only
 (b) A and C only
 (c) A, C and D only
 (d) B, C and D only

Answer: (b)

Question: The correct order of stability of following diazonium ions is



Options:

- (a) $a < b < c < d$
- (b) $a < b < d < c$
- (c) $c < d < b < a$
- (d) $d < c < b < a$

Answer: (a)

Question: $\text{K}_2\text{Cr}_2\text{O}_7 + \text{I}^- + \text{H}^+ \rightarrow \text{I}_2$ (x = number of moles of e^- exchanged per mol I_2)
 $\text{K}_2\text{Cr}_2\text{O}_7 + \text{S}^{2-} \rightarrow \text{S}$ (y = number of moles of e^- exchanged for mole of S) $x + y$ is

Options:

- (a) 12
- (b) 9
- (c) 4
- (d) 6

Answer: (c)

Question: Match the column

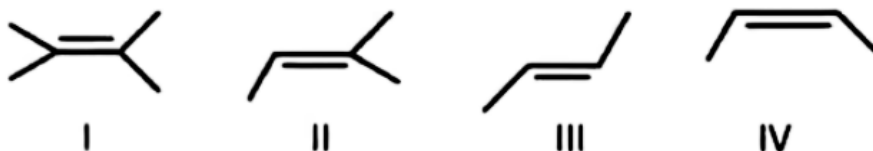
Column-I	Column-II
A) IF_3	i) sp^3d^3 , Pentagonal bipyramidal
B) IF_5	ii) sp^3d^3 , T-shaped
C) IF_7	iii) sp^3 , Tetrahedral
D) ClO_4^-	iv) sp^3d^2 , Square pyramidal

Options:

- (a) A-(i), B-(ii), C-(iii), D-(iv)
- (b) A-(ii), B-(i), C-(iv), D-(iii)
- (c) A-(ii), B-(iv), C-(i), D-(iii)
- (d) A-(ii), B-(iii), C-(iv), D-(i)

Answer: (c)

Question: Consider the following alkene



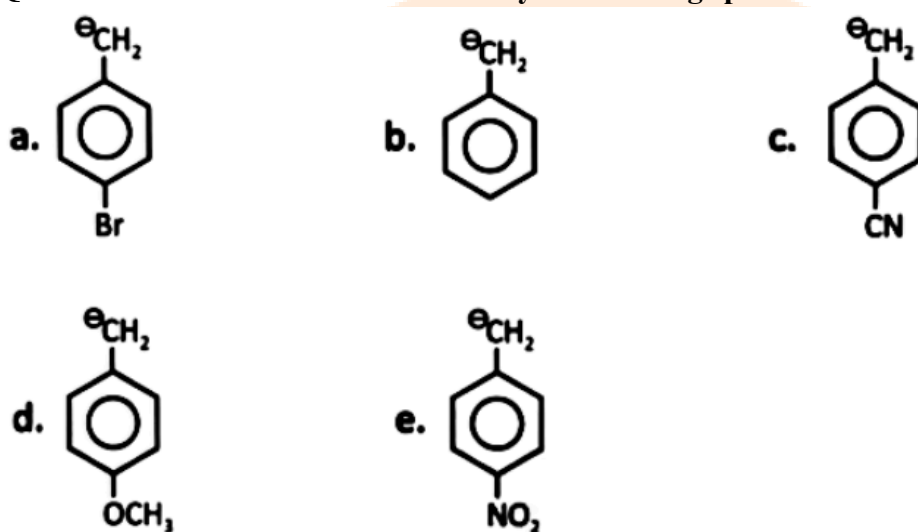
The correct stability order of alkenes is

Options:

- (a) II > I > III > IV
- (b) I > II > IV > III
- (c) I > II > III > IV
- (d) III > I > II > IV

Answer: (c)

Question: The correct order of stability of following species is

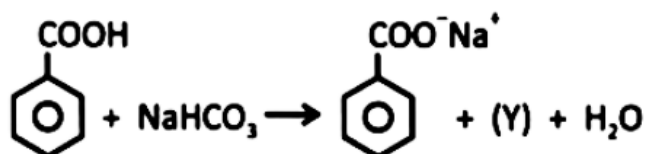
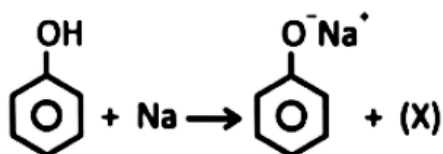


Options:

- (a) e > c > a > b > d
- (b) d > c > b > a > e
- (c) e > a > c > b > d
- (d) e > a > b > c > d

Answer: (a)

Question: What is the sum of molar mass of X and Y formed in the given reactions?



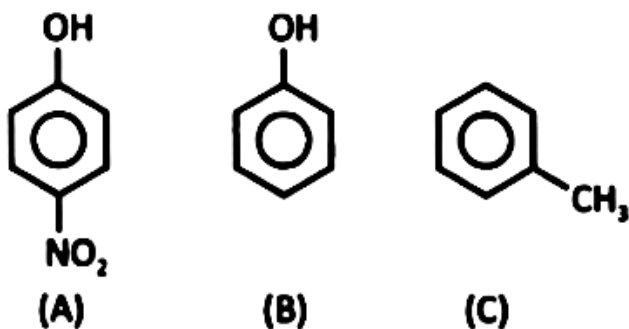
Options:

- (a) 46
- (b) 44
- (c) 2

(d) 42

Answer: (a)

Question: Consider the following molecules.



The correct order of dipole moment is

Options:

- (a) $A > B > C$
- (b) $A > C > B$
- (c) $B > A > C$
- (d) $C > A > B$

Answer: (a)

Question: Given below are two statements.

Statement-I: Atomic radius is always more than ionic radius.

Statement-II: The correct order of metallic character is $K > Mg > Al > B$

In the light of above statements, choose the correct option.

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both Statement-I and Statement-II are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (d)

Question: Match the following

Column-I	Column-I
A) Free expansion	i) $W = -P_{\text{ex}}\Delta V$
B) Reversible isothermal	ii) $W = nC_v dT$
C) Irreversible isothermal	iii) $W = 0$
D) Adiabatic reversible	iv) $W = -nRT \ln \frac{V_r}{V_1}$

Options:

- (a) A(i), B(iv), C(iii), D(ii)
- (b) A(iii), B(iv), C(i), D(ii)
- (c) A(iv), B(iii), C(ii), D(i)
- (d) A(ii), B(i), C(iii), D(iv)

Answer: (b)

Question: Non-volatile solute A of mass 0.3 g (Molecular mass = 60), and non-volatile solute B of mass 0.9 g (Molecular mass = 180) in 100 mL H_2O at $27^\circ C$. If $K_b = 0.52 K \cdot Kg \cdot mol^{-1}$, then elevation of boiling point is

Options:

- (a) 0.52 K
- (b) 0.052 K
- (c) 0.026 K
- (d) 0.083 K

Answer: (b)

Question: A solution contains two group-IV cations, X^{2+} and Y^{2+} , each at an initial concentration of 0.1 M. H_2S gas is passed through the solution to form a saturated solution. Given

$$K_{sp} \text{ of } YS = 2 \times 10^{-27} M^2$$

$$K_{sp} \text{ of } XS = 1 \times 10^{-27} M^2$$

What is the minimum concentration of sulphide in $[S^{2-}]$ required to begin precipitation of YS?

Options:

- (a) 2×10^{-26}
- (b) 10^{-26}
- (c) 3.2×10^{-14}
- (d) 0.1

Answer: (a)

Question: Two solutes A and B of 0.3 g and 0.9 g respectively (molar mass of A and B are 30 g/mol and 90 g/mol respectively). Calculate of osmotic pressure at 300 K (in atm)

Options:

Answer: (5)

Question: Minimum energy transition of Balmer series (energy line having minimum energy) of H-atom has energy of L eV. If the value of minimum energy of Lyman series (energy line having minimum energy) of H-atom in terms of L is y, then the value of 10y is

Options:

Answer: (2)

Question: Find % of 'N' in 0.5 g organic compound which gives 34 mL $N_2(g)$ at 715 mm Hg pressure and 300 K Aq. tension = 15 mm Hg

(Report to nearest integer) $R = 0.0821 \frac{Lit-atm}{K-mol}$

Options:

Answer: (7)

Question: Find the value of $\log\left(\frac{K_{catalyst}}{K_{uncatalyst}}\right)$ at 300K. If the change in activation energy

(DEa) is -10 kJ/mol.

($R = 8 \text{ J K}^{-1} \text{ mol}^{-1}$) ($\ln x = 2.31 \log x$)

Options:

Answer: (2)



JEE-Main-24-01-2026 (Memory Based)

[MORNING SHIFT]

Maths

Question: The value of $\frac{\sqrt{3}\operatorname{cosec}20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}$ is

Options:

- (a) 12
- (b) 16
- (c) 64
- (d) 32

Answer: (c)

$$\begin{aligned} & \frac{\sqrt{3}\operatorname{cosec}20^\circ - \sec 20^\circ}{\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ} \\ &= \frac{2\left(\frac{\sqrt{3}}{2}\cos 20^\circ - \sin 20^\circ\right)}{\sin 20^\circ \cos 20^\circ} \\ &= 2 \times 2 \frac{\sin(20^\circ - 20^\circ)}{\frac{\sin 40^\circ \cdot \sin 20^\circ}{2 \cos 20^\circ \times \frac{1}{2}}} \\ &= \frac{4}{\frac{1}{2} \sin 20^\circ} = 4 \times 2^4 = 64 \end{aligned}$$

Question: The number of solution for $x \in \mathbb{R}$, $x|x-4|+|x-1|-2=0$

Options:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Answer: (a)

$C - I$

$$-x(x-4) - (x-1) - 2 = 0$$

$$-x^2 + 4x - x + 1 - 2 = 0$$

$$-x^2 + 3x - 1 = 0 \Rightarrow x^2 - 3x + 1 = 0$$

1 real root

$C - II$

$$1 < x < 4$$

$$-x(x-4) + (x-1) - 2 = 0$$

$$-x^2 + 4x + x - 3 = 0$$

$$-x^2 + 5x - 3 = 0$$

$$+x^2 - 5x + 3 = 0$$

No real root

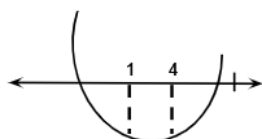
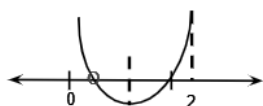
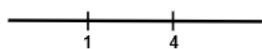
$C - III$

$$x > 4$$

$$x(x-4) + x - 1 - 2 = 0$$

$$x^2 - 3x - 3 = 0$$

No real root



Question: If $\cot x = \frac{5}{12}$ for some $x \in \left(\pi, \frac{3\pi}{2}\right)$ then $\sin 7x \left(\cos \frac{13x}{2} + \sin \frac{13x}{2} \right) + \cos 7x \left(\cos \frac{13x}{2} - \sin \frac{13x}{2} \right)$ is equal to

Answer: $\left(\frac{1}{\sqrt{13}}\right)$

$$\sin 7x \left(\cos \frac{13x}{2} - \cos 7x \sin \frac{13x}{2} \right) + \sin 7x \sin \frac{13x}{2} +$$

$$\sin \left(7x - \frac{13x}{2} \right) + \cos \left(7x - \frac{13x}{2} \right)$$

$$= \sin \frac{x}{2} + \cos \frac{x}{2} = \sqrt{1 + 2 \sin \frac{x}{2} \cos x}$$

$$\sqrt{1 - \sin x}$$

$$= \sqrt{1 - \frac{12}{13}}$$

$$= \frac{1}{\sqrt{13}}$$

Question: If the function $f(x) = \frac{e^x (e^{\tan x - x} - 1) + \log_e (\sec x + \tan x) - x}{\tan x - x}$

0, then the value of $f(0)$ is equal to

Options:

(a) $\frac{1}{2}$

(b) 2

(c) $\frac{2}{3}$

is continuous at $x =$

(d) $\frac{3}{2}$

Answer: (d)

$$\begin{aligned}
 F(0) &= \lim_{x \rightarrow 0} \frac{e^x(e^{\tan x - x} - 1)}{\tan x - x} + \frac{\ln(\sec x + \tan x) - x}{\tan x - x} \\
 &= e^0(1) = \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{\tan x - x} \\
 &= 1(1) + \lim_{x \rightarrow 0} \frac{\sec x - 1}{\sec^2 x - 1} \\
 &= 1 + \lim_{x \rightarrow 0} \frac{(\sec x - 1)}{(\sec x - 1)(\sec x + 1)} \\
 &= 1 + \lim_{x \rightarrow 0} \frac{1}{(\sec x + 1)} \\
 &= 1 + \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

Question: Consider 10 data such that their mean is 10 and variance is 2. If one of the data α is removed and new data entry β is inserted. Now new mean is 10.1 and new variance is 1.99 then $(\alpha + \beta)$ is equal to

Options:

- (a) 10
- (b) 20
- (c) 1
- (d) 2

Answer: (c)

$$\begin{aligned}
 \sum x^2 &= 102 \times 10 \\
 \frac{0^2 \sum x_1^2 - \alpha^2 + \beta^2}{10} (10.1)^2 &= 1.99 \\
 \frac{1020 - \alpha^2 + \beta^2}{10} &= 1.99 + 1020 \\
 1020 - \alpha^2 + \beta^2 &= 1040 \\
 \beta^2 - \alpha^2 &= 20 \\
 (\beta - \alpha)(\beta + \alpha) &= 20 \\
 1 \times \beta + \alpha &= 20 \\
 \beta + \alpha &= 20
 \end{aligned}$$

Question: Consider an A.P $a_1, a_2, \dots, a_n$; $a_1 > 0$, $a_2 - a_1 = -\frac{3}{4}$, $a_n = \frac{1}{4}$ and $\sum_{i=1}^n a_i = \frac{525}{2}$ then

$\sum_{i=1}^{17} a_i$ is equal to

Answer: (136)

$$a_1 - a_1 = d = -\frac{3}{4}$$

$$a + (n-1)\left(-\frac{3}{4}\right) = \frac{1}{4}9$$

$$\Rightarrow \frac{3}{4}a = (n-1)\frac{3}{4} \Rightarrow a = n-1 \quad n = a+1$$

$$\frac{n}{2} \left[2a + (n-1)\left(-\frac{3}{4}\right) \right] = \frac{525}{2}$$

$$(a+1) \left[2a - \frac{39}{4} \right] = 525$$

$$(s+1) \times \frac{59}{2} = 525$$

$$(a^2 + a) = (105) \times 2$$

$$a(a+1) = 210 \Rightarrow a = 14$$

$$d = -\frac{3}{4}$$

$$\sum_{i=1}^{17} a_i = \frac{17}{2} \left[28 + 16 \times -\frac{3}{4} \right]$$

$$= \frac{17}{2} [28 - 12] = \frac{17}{2} \times 16 = 17 \times 8$$

$$= 136$$

Question: If $F(t) = \int \frac{1 - \sin(\ln t)}{1 - \cos(\ln t)} dt$ and $F(e^{\pi/2}) = -e^{\pi/2}$ then $F(e^{\pi/4})$ is

Options:

(a) $(-1 - \sqrt{2})e^{\frac{\pi}{4}}$

(b) $(1 - \sqrt{2})e^{\frac{\pi}{4}}$

(c) $(1 + \sqrt{2})e^{\frac{\pi}{4}}$

(d) $(-2 - \sqrt{2})e^{\frac{\pi}{4}}$

Answer: (a)

$$f(t) = \int \frac{1 - \sin \ln t}{1 - \cos \ln t}$$

$$\int \frac{1 - \sin x}{1 - \cos x} e^x$$

$$\left(\frac{1}{2} \operatorname{cosec}^2 \frac{x}{2} - \cot \frac{x}{2} \right) e^x$$

$$f(t) = e^x \left(\cot \frac{x}{2} \right) + C$$

$$f(t) = \frac{\cot \ln t}{2} e^{\ln t} + C$$

$$f(t) = -e^{\ln t} \frac{\cot \ln t}{2} + C$$

$$f(t) = \frac{\cot \ln t}{2} + C$$

$$C = 0$$

$$f(e^{\frac{\pi}{4}}) = e^{\frac{\pi}{4}} \cot \frac{\pi}{8}$$

$$= -e^{\frac{\pi}{4}} (1 + \sqrt{2})$$

Question: Find domain of $\log_{(10x^2-17x-7)} (18x^2 - 11x + 1)$ is

Answer: $(-\frac{6}{5})$

$$18x^2 - 11x + 1 > 0$$

$$(2x - 1)(9x - 1) > 0$$

$$x \in (-\infty, \frac{1}{9}) \cup (\frac{1}{2}, \infty)$$

$$10x^2 - 17x + 7 > 0$$

$$10x^2 - 10x - 7x + 7 > 0$$

$$10x(x - 1) - 7(x - 1) > 0$$

$$(x - 1)(10x - 7) > 0$$

$$x \in (-\infty, \frac{7}{10}) \cup (1, \infty)$$

$$\text{and } 10x^2 - 17x + 7 \neq 1$$

$$10x^2 - 17x + 6 \neq 0$$

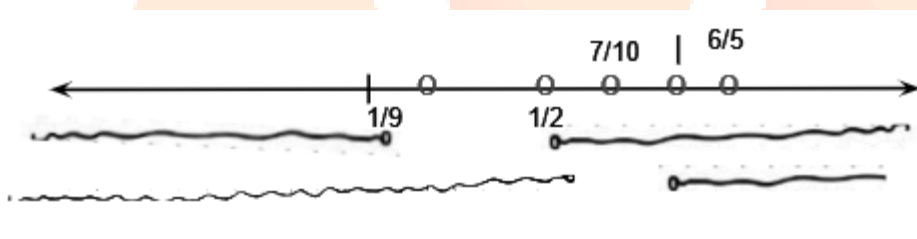
$$10x^2 - 12x - 5x + 6 \neq 0$$

$$2x(5x - 6) - 1(5x - 6) \neq 0$$

$$(5x - 6)(2x - 1) \neq 0$$

$$x \neq \frac{1}{2}, \frac{6}{5}$$

$$(-\infty, \frac{1}{9}) \cup (\frac{1}{2}, \frac{7}{10}) \cup (1, \infty) - \{\frac{6}{5}\}$$



Question: $\frac{1}{25!} + \frac{1}{3!23!} + \dots$ upto 13 terms = ?

Answer: $(\frac{1}{26!} \times 2^{25})$

$$\frac{1}{25!x!} + \frac{1}{8!23!} \dots \dots \dots \text{upto 13}$$

$$T_n = \frac{1}{(2n-1)!(27-2n)!}$$

$$T_n = \frac{(26!)}{26! \times 2! \times \{(2n-1)! \times (27-2n)!\}}$$

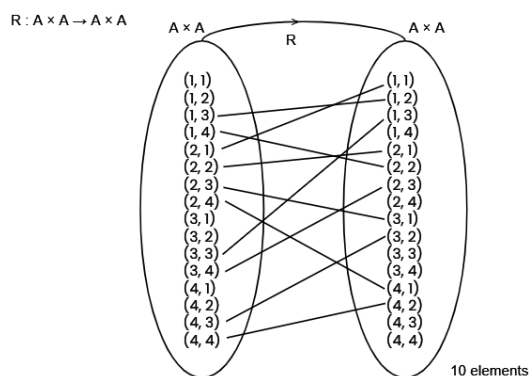
$$= \frac{1}{26!} {}^{26}C_{2n-1}$$

$$\frac{1}{26!} [{}^{26}C_1 + {}^{26}C_3 \dots \dots \dots]$$

$$\frac{1}{26!} \times 2^{25}$$

Question: If $A \in \{1, 2, 3, 4\}$; A relation defined on $A \times A$ such that $2a + 3b = 3c + 4d$ then number of elements in relation

Answer: (10 elements)



Question: Number of 4 - digit number by using digits 0, 1, 2, 5, 9 divisible by 3 & number lies between 5000 to 9000 (Repetition is allowed)
Answer: (42)

$$\overline{[0, 1, 2, 3, \text{ and } 9]} \quad [\quad] \quad [\quad]$$

$$= \text{Sum of coefficient of } x^{3k} \text{ in } x^5 (x^0 + x^1 + x^2 + x^5 + x^9)^3$$

$$= \frac{5^3 + \omega^5 (-\omega)^3 + (-\omega^2)^3 \omega^{10}}{3} = 42$$

Question: $\left| \frac{z-6i}{Z-2i} \right| = 1$ & $\left| \frac{z-8+2i}{z+2i} \right| = \frac{3}{5}$ If Z is complex no satisfying then find $\sum |2|^2$

Answer: (385)

$$\left| \frac{z-6i}{z-2i} \right| = 1$$

$$\left| \frac{z-(-2i)}{z-(8-2i)} \right| = \frac{5}{3}$$

This is equal of circle

$$\text{Radins} = \frac{\frac{5}{3}}{\left(\frac{5}{3}\right)^2 - 1} AB$$

$$= \frac{\frac{5}{3} \times 9}{25-9} \times 8 = \frac{15}{16} \times 8 = \frac{15}{2}$$

Create $\rightarrow$ mid point of I and E

$$\left(\frac{29}{2}, -2 \right)$$

eqⁿ of circle

$$\left(x - \frac{25}{2} \right)^2 + (y+2)^2 = \frac{225}{4}$$

$$z_i = 8 + 4i$$

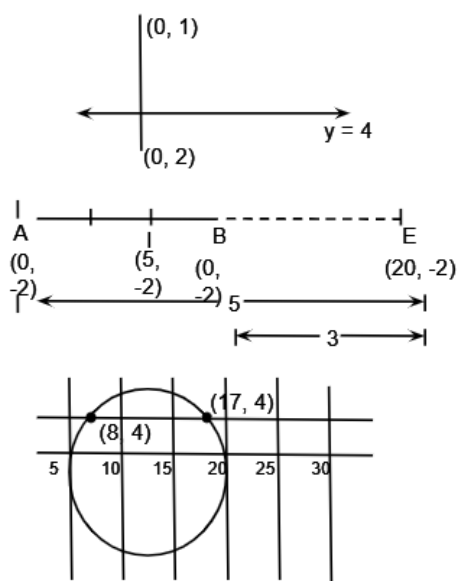
$$z_2 = 17 + 4i$$

$$|z_i|^2 + |z_2|^2$$

$$64 + 16 + 289 + 16$$

$$80 + 305$$

$$= 385$$



Question: Consider a sequence 729, 81, 9, 1... Let P_n = product of first n terms of the given sequence $\sum_{n=1}^{40} (P_n)^{\frac{1}{n}} = \frac{3\alpha - 1}{2 \times 3^\beta}$. and Then the value of $\alpha + \beta$ is

Options:

- (a) 73
- (b) 75
- (c) 76
- (d) 81

Answer: (a)

$$p_n = a_1, a_2, a_3, \dots, a_{27}$$

$$= 3^{8-2 \times 1} \cdot 3^{8-2 \times 2} \cdot 3^{8-2 \times 3} \dots e^{2^{8-2n}}$$

$$p_n = 3^{8-8+8-4} \dots 8-2n$$

$$= 3^{6+4+2} \dots 8-2n$$

$$3^{\frac{n}{2}} (6 + 8 - 2n)$$

$$p_n \frac{1}{n} 3^{\frac{n}{2}} (7 - n) = 3^{n(7-n)}$$

$$= 3^{n-n}$$

$$\sum_n^{40} 37n = 3^6 + 3^5 \dots - 3^7 - 40$$

$$= 36 \frac{(1-(1/3)^{40})}{(1-\frac{1}{3})}$$

$$= 36 \times \frac{3}{2} \cdot (1 - 3^{-40})$$

$$729, 81, 9, 1$$

$$a = 36$$

$$r = \frac{34}{36} = 3^{-2}$$

$$ak = 729 \cdot (3^{-1})^k - 1$$

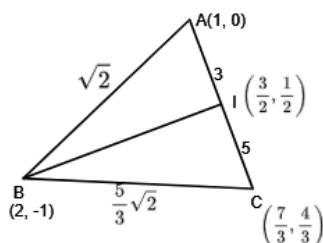
$$= 729 \cdot 3^{-2k+2}$$

$$= 3^6 \cdot 3^{-2k+2}$$

$$= 3^{-2k+8}$$

Question: $(1,0)$, $(2,-1)$, $\left(\frac{7}{3}, \frac{4}{3}\right)$ are vertices of a Δ . Find eqⁿ of internal angle bisector through $(2, -1)$,

Answer: $(3x + y = 5)$



$$BC = \sqrt{\left(\frac{7}{3} - 2\right)^2 + \left(\frac{4}{3} + 1\right)^2}$$

$$= \sqrt{\frac{1}{9} + \frac{49}{9}}$$

$$I\left(\frac{7+5}{8}, \frac{4+0}{8}\right) = \left(\frac{3}{2}, \frac{1}{2}\right)$$

$$m_{BI} = \frac{\frac{1}{2} + 1}{\frac{3}{2} - 2} = \frac{3}{-1} = -3$$

$$\text{Eq}^n \text{ of } BI \Rightarrow y + 1 = -3(x - 2)$$

$$y + 1 = -3x + 6$$

$$3x + y = 5$$

Question: Number of matrices A of order 3×2 such that all of its elements are from the set $\{-2, -1, 0, 1, 2\}$ such that trace of AA^T is 5, is equal to

Options:

- (a) 120
- (b) 312
- (c) 192
- (d) 126

Answer: (a)

$$\begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \\ c_1 & c_2 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} = \begin{matrix} a_1^2 + a_2^2 & & \\ & b_1^2 + b_2^2 & \\ & & c_1^2 + c_2^2 \end{matrix}$$

$$[a_1^2 + a_2^2 + b_1^2 + b_2^2 + c_1^2 + c_2^2] = 5$$

$${}^6C_2 \times 2!(-2) = 120$$

Question: Out of 100 bulbs, 10 are defective and 90 are non defective. If the probability of finding 7 defective bulbs out of 8 draws, with replacement, is $\frac{K}{10^8}$ then the value of K is

Options:

- (a) 69
- (b) 72

(c) 75

(d) 96

Answer: (b)

$$P = \frac{10}{100} = \frac{1}{10} \quad q = \frac{9}{10}$$

$$p(x = r) = {}^n C_r p^r q^{n-r}$$

$$P(x = 7) = {}^8 C_7 \left(\frac{1}{10}\right)^7 \left(\frac{9}{10}\right)^1$$

$$= \frac{8 \times 1}{10^8} \times 9 = \frac{72}{10^8}$$

$$k = 72$$

Question: Let the lines $L_i : \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}), \lambda \in R$

Intersect at the point R. Let P and Q be the points lying on lines L_1 & L_2 respectively

such that $|PR| = \sqrt{29}$ and $|PQ| = \sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(QR)^2$ is equal to

Options:

(a) 340

(b) 360

(c) 320

(d) 348

Answer: (b)

$$L_1 : \vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$L_2 : \vec{r} = (4\hat{i} + \hat{j}) + \mu(5\hat{i} + 2\hat{j} + \hat{k})$$

$$P(1 + 2\lambda, 2 + 3\lambda, 3 + 4\lambda)$$

$$Q(4 + 5\mu, 1 + 2\mu, 1 + \mu)$$

$$R(-1, -1, -1)$$

$$(2\lambda + 2)^2 + (3 + 3\lambda)^2 + (4\lambda + 3)^2$$

$$R(1 + 2\lambda + 9 + 5\mu)$$

$$3\pi 5\mu - 2\lambda = -3$$

$$(2\lambda + 2)^2 + (3 + 3\lambda)^2 + (4\lambda + \lambda)^2$$

$$= 2 \times 2\mu - 3\lambda = 1 - 11$$

$$29(d+1)^2 = 29$$

$$\lambda = -1 \cdot 2 + 1 = -1$$

$$\lambda = 0, \lambda = ?$$

$$P(1, 2, 3)$$

$$PQ = (5+3)^2(24-1)^2 + (\mu-3)^2 = \frac{47}{3}$$

$$P(1, 2, 3) Q\left(\frac{7}{3}, \frac{1}{3}, \frac{1}{3}\right) = \mu^2 - 6\mu + 9$$

$$30\mu^2 + 20\mu + 19 = \frac{47}{3}$$

$$(QR)^2 = \frac{100}{9} + \frac{16}{9} + \frac{4}{9} = \frac{120}{9} \quad 9\mu^2 + 6\mu + 1 = 0$$

$$= \frac{40}{3} \Rightarrow 27(QR)^2 = 27 \times \frac{40}{3} = 360$$

Question: If $\int_0^{36} \left(\frac{tx}{36}\right) dt = 4\alpha f(x), y = f(x)$ is standard parabola passing through (2, 1) & (-4, β). Find the value of $\beta^\alpha =$

Answer: (64)

$$y = kx^2 \Rightarrow 1 = k(2)^2 \Rightarrow k = \frac{1}{4}$$

$$f(x) = y = \frac{1}{4}x^2 \quad \beta = \frac{1}{4}(-4)^2 = 4$$

$$\int_0^{36} \frac{1}{4} \left(\frac{tx}{36}\right)^2 dt = 4\alpha \times \frac{1}{4}x^2$$

$$\frac{1}{3} \times \frac{x^2}{(36)^2 \times 4} (t^3)_0^{36} = \alpha x^2$$

$$\frac{1}{3} \times \frac{1}{4 \times (36)^2} (36)^3 = \alpha$$

$$\frac{1}{3} \times \frac{36}{4} = \alpha \Rightarrow \alpha = 3$$

$$\beta^\alpha = 4^3 = 64$$

Question: A_1 is the area bounded by $y = x^2 + 1$, $x + y = 8$, y-axis in the 1st quadrant and A_2 is the area bounded by $y = x^2 + 2$, $y^2 = x$, $x = 0$ and $x = 2$ in the 1st quadrant find ($A_1 - A_2$)

Options:

(a) $\frac{2}{3} + \frac{4\sqrt{2}}{3}$

(b) $\frac{3}{2} + \frac{4\sqrt{2}}{3}$

(c) $\frac{3}{5} + \frac{4\sqrt{2}}{3}$

(d) None of these

Answer: (a)

$$\begin{aligned}
 A &= \int_0^2 (8-x) - (x^2+2) \\
 &= \int_0^2 (8-x-x^2) \\
 &= \left\{ 6x - \frac{x^2}{2} - \frac{x^3}{3} \right\}_{-3}^2 = \left(12 - 2 - \frac{8}{3} \right) = \left(\frac{22}{3} \right) \\
 &= 1 \\
 \int_0^2 (x^2+2-\sqrt{x}) &= \left[\frac{x^3}{3} + 2x - \frac{2}{3}x\frac{3}{2} \right]_0^2 \\
 &= \left(\frac{8}{3} + 4 - \frac{2}{3}2\sqrt{2} \right) \\
 A_1 - A_2 &= \frac{22}{3} - \left(\frac{20}{3} - \frac{4\sqrt{2}}{3} \right) \\
 &= \left(\frac{20}{3} - \frac{4\sqrt{2}}{3} \right) \\
 &= \frac{2}{3} + \frac{4\sqrt{2}}{3}
 \end{aligned}$$

Question: Let a circle pass through points $A(-\sqrt{2}\alpha, 0)$, $B(0, \sqrt{3}\beta)$ And $O(0,0)$ such that its radius is 4. Then the radius of locus of centroid of triangle OAB is

Options:

- (a) $\frac{2}{3}$
- (b) $\frac{8}{3}$
- (c) $\frac{4}{3}$
- (d) $\frac{11}{3}$

Answer: (b)

$$OC = R\sqrt{\frac{\alpha^2}{2} + \frac{3}{4}\beta^2} = 4$$

$$2\alpha^2 + 3\beta^2 = 64$$

$$x = -\frac{\sqrt{2}\alpha}{3}, \quad y = \frac{\sqrt{3}\beta}{3}$$

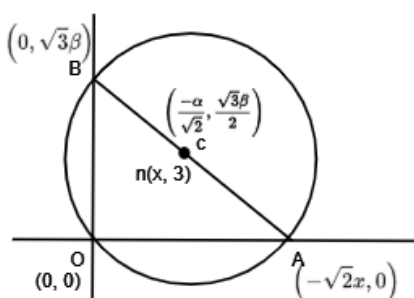
$$3x = -\sqrt{2}\alpha \quad 3y = \sqrt{3}\beta$$

$$\Rightarrow 9x^2 = 2\alpha^2$$

$$9x^2 + 9y^2 = 64$$

$$x^2 + y^2 = \frac{64}{9}$$

$$\text{Radius} = \frac{8}{3}$$



$$\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{b} = \hat{i} - \hat{j}$$

$$\vec{c} = \vec{a} \times \vec{b}$$

Question: $\vec{d}$ makes an angle $\frac{\pi}{4}$ with $\vec{C}$ and $|\vec{c} \times \vec{d}| = 3, |\vec{a} - \vec{d}| = \sqrt{11}$. Find the value of

Answer: $(-\frac{1}{2})$

$$\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{b} = \hat{i} - \hat{j}$$

$$\vec{c} = \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 1 & -1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} - 4\hat{k}$$

$$|\vec{c}| = \sqrt{18} = 3\sqrt{2}$$

$$|\vec{c} \times \vec{d}| = |\vec{c}||\vec{d}| \sin \frac{\pi}{4}$$

$$3 = \left(\frac{3}{\sqrt{2}}\right)|\vec{d}| \times \frac{1}{\sqrt{2}} \Rightarrow |\vec{d}| = 1$$

$$|\vec{a} - \vec{d}| = \sqrt{11} \Rightarrow |\vec{a}|^2 + |\vec{d}|^2 - 2(\vec{a} \cdot \vec{d}) = 11$$

$$= 9 + 1 - 2(\vec{a} \cdot \vec{d}) = 11$$

$$= \vec{a} \cdot \vec{d} = -\frac{1}{2}$$